

Movie Rating Prediction

Internship Report - Month 2

Task 4

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1. Introduction

Movies are an important form of entertainment, and audience ratings play a significant role in understanding how movies are received by viewers. IMDb ratings provide a numerical representation of audience opinions and can be influenced by several factors such as movie genre, duration, metascore, number of votes, and reviews.

The **Movie Rating Prediction** project applies data analysis and machine learning techniques to predict IMDb movie ratings based on available movie characteristics. The project uses a dataset containing information about **10,000 movies**, including their year, genre, duration, certificate, metascore, votes, review count, and ratings.

The dataset is first cleaned and prepared by handling missing values and converting text-based numerical information into usable formats. Exploratory Data Analysis is then performed to identify patterns and relationships within the data. Finally, multiple regression models are trained and evaluated to determine which model provides the most accurate rating predictions.

The main goal of this project is to demonstrate the practical application of **data preprocessing, exploratory data analysis, feature engineering, machine learning regression, and model evaluation** in a real-world movie rating prediction problem.

2. Project Objective

The main objectives of this project are:

- To predict movie ratings using machine learning regression techniques.
- To analyze movie-related features and their relationship with IMDb ratings.
- To clean and preprocess the movie dataset for machine learning.
- To handle missing values and convert text-based numerical columns into numeric format.
- To perform Exploratory Data Analysis (EDA) to understand movie rating patterns.
- To select relevant numerical and categorical features for prediction.
- To prepare the data for machine learning using feature preprocessing techniques.
- To build and evaluate Linear Regression, Decision Tree Regression, and Random Forest Regression models.
- To compare the performance of the developed regression models using MAE, RMSE, and R^2 Score.
- To identify the best-performing model for predicting movie ratings.

3. Project Overview

The Movie Rating Prediction project focuses on using machine learning techniques to predict IMDb movie ratings based on various movie-related features.

The project uses a dataset containing information about **9,596 movies**, including year, duration, genre, certificate, metascore, number of votes, review count, and IMDb rating.

The dataset was first cleaned and preprocessed by handling missing values and converting text-based numerical fields into appropriate numeric formats. Exploratory Data Analysis (EDA) was then performed to understand rating distributions, relationships between numerical features, and average ratings across different genres.

Three regression models were developed and evaluated:

- **Linear Regression**
- **Decision Tree Regression**
- **Random Forest Regression**

The models were evaluated using **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **R² Score**. Among the three models, **Random Forest Regression** achieved the best overall performance, with an **MAE of 0.512**, **RMSE of 0.709**, and **R² Score of 0.539**.

The project demonstrates how machine learning can be applied to analyze movie characteristics and predict ratings while also highlighting the limitations of prediction when only a selected set of movie features is available.

4. Dataset Description

The dataset used in this project is an IMDb Movies Dataset containing information about 10,000 movies. It provides several attributes related to movies, including their ratings, year, duration, genre, metascore, votes, and reviews.

The dataset contains the following main columns:

Column	Description
Poster	Movie poster information
Title	Name of the movie
Year	Release year of the movie
Certificate	Movie certification
Duration (min)	Movie duration in minutes
Genre	Genre or genres associated with the movie
Rating	IMDb movie rating
Metascore	Critic score of the movie
Director	Director of the movie
Cast	Main cast information
Votes	Number of IMDb votes
Description	Movie description
Review Count	Number of reviews
Review Title	Title of the review
Review	Review information

For the prediction task, the Rating column was selected as the target variable. The following features were selected as relevant inputs for the machine learning models:

- Year
- Duration (min)
- Genre
- Certificate
- Metascore
- Votes
- Review Count

After removing records with missing target ratings, 9,596 movie records remained for the machine learning process.

5. Tools and Technologies Used

The following tools and technologies were used to develop the Movie Rating Prediction project:

Programming Language

- **Python** – Used for data preprocessing, exploratory data analysis, machine learning model development, and evaluation.

Development Environment

- **Jupyter Notebook** – Used to write, execute, and document the project code.

Libraries

- **Pandas** – Used for data loading, data manipulation, and preprocessing.
- **NumPy** – Used for numerical operations and calculations.
- **Matplotlib** – Used for creating data visualizations.
- **Seaborn** – Used for statistical visualizations and correlation analysis.
- **Scikit-learn** – Used for data preprocessing, train-test splitting, machine learning models, and model evaluation.

Machine Learning Techniques

- Linear Regression
- Decision Tree Regression
- Random Forest Regression

Evaluation Metrics

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

6. Project Methodology

The Movie Rating Prediction project follows a structured machine learning workflow. The major steps performed in the project are:

6.1 Loading the Dataset

The IMDb Movies Dataset was loaded into a Pandas DataFrame for analysis and processing.

6.2 Dataset Inspection

The dataset was inspected using its shape, information, descriptive statistics, column names, missing values, and duplicate records to understand its structure and quality.

6.3 Data Preprocessing

The data was prepared for machine learning by removing records with missing target ratings, converting text-based numerical columns into numeric format, and handling remaining missing values.

6.4 Feature Selection

Relevant numerical and categorical features were selected for predicting movie ratings. Text-heavy and identification-related columns were excluded from the initial regression models.

6.5 Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the distribution of movie ratings and relationships between important movie features. Rating distribution, rating versus metascore, rating versus number of votes, average rating by genre, and correlation analysis were examined.

6.6 Preparing Data for Machine Learning

The processed dataset was divided into input features and the target variable. The data was then split into training and testing sets.

6.7 Feature Preprocessing

Numerical features were standardized using **StandardScaler**, while categorical features were converted into numerical form using one-hot encoding.

6.8 Model Building

Three regression models were developed:

- Linear Regression
- Decision Tree Regression
- Random Forest Regression

6.9 Model Evaluation

The models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

6.10 Best Model Analysis

The model performance was compared to identify the best-performing model. Random Forest Regression achieved the best overall results and was selected for further prediction analysis.

6.11 Actual vs. Predicted Analysis

The actual IMDb ratings were compared with the ratings predicted by the Random Forest model using a scatter plot.

7. Code Explanation

The project was implemented in Python using Jupyter Notebook. The following code sections explain the main implementation steps used for movie rating prediction.

7.1 Importing Libraries

The required Python libraries were imported for data manipulation, visualization, preprocessing, machine learning, and model evaluation.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
```

```
from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
```

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

7.2 Loading the Dataset

The IMDb Movies Dataset was loaded using Pandas.

```
df = pd.read_csv("imdb-movies-dataset.csv")

df.head()
```

7.3 Dataset Inspection

The dataset was inspected to understand its structure, columns, statistical information, missing values, and duplicate records.

```
df.shape
df.info()
df.columns
df.describe()
df.isnull().sum()
df.duplicated().sum()
```

7.4 Removing Missing Target Values

Records with missing IMDb ratings were removed because the "Rating" column is the target variable for prediction.

```
df = df.dropna(subset=['Rating'])
```

```
df.shape
```

After this step, 9,596 records remained.

7.5 Converting Numerical Columns

The "Votes" and "Review Count" columns were stored as text values. Commas were removed and the columns were converted into numeric format.

```
df['Votes'] = pd.to_numeric(
    df['Votes'].str.replace(',', '', regex=False),
    errors='coerce'
)
```

```
df['Review Count'] = pd.to_numeric(
```

```
df['Review Count'].str.replace(',', '', regex=False),  
errors='coerce'  
)
```

7.6 Rechecking Missing Values

After converting the text-based numerical columns, the missing values were checked again to identify any additional missing values created during numeric conversion.

```
df.isnull().sum()
```

7.7 Selecting Relevant Features

Relevant numerical and categorical features were selected for predicting movie ratings. Text-heavy and identification-related columns were excluded from the initial regression models.

```
features = [  
    'Year',  
    'Duration (min)',  
    'Genre',  
    'Certificate',  
    'Metascore',  
    'Votes',  
    'Review Count'  
]
```

```
target = 'Rating'
```

```
model_df = df[features + [target]].copy()
```

```
model_df.head()
```

7.8 Handling Missing Values

Missing values in numerical features were handled using the median, while missing categorical values were replaced with the most frequent category.

```
numeric_features = [  
    'Year',  
    'Duration (min)',  
    'Metascore',  
    'Votes',  
    'Review Count'  
]  
  
categorical_features = [  
    'Genre',  
    'Certificate'  
]  
  
for column in numeric_features:  
    model_df[column] = model_df[column].fillna(  
        model_df[column].median()  
    )  
  
for column in categorical_features:  
    model_df[column] = model_df[column].fillna(  
        model_df[column].mode()[0]  
    )  
  
model_df.isnull().sum()
```

After preprocessing, all selected features contained no missing values.

7.9 Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the distribution of movie ratings and the relationships between important movie features.

7.9.1 Distribution of Movie Ratings

The distribution of movie ratings was visualized using a histogram with a KDE curve.

```
plt.figure(figsize=(8, 5))

sns.histplot(
    model_df['Rating'],
    bins=20,
    kde=True
)

plt.title('Distribution of Movie Ratings')
plt.xlabel('Rating')
plt.ylabel('Number of Movies')
plt.show()
```

7.9.2 Rating vs. Metascore

A scatter plot was used to examine the relationship between IMDb ratings and Metascores.

```
plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=model_df,
    x='Metascore',
    y='Rating',
    alpha=0.5
)

plt.title('Movie Rating vs. Metascore')
plt.xlabel('Metascore')
plt.ylabel('IMDb Rating')
plt.show()
```

These visualizations help identify the distribution of movie ratings and whether higher critic scores are generally associated with higher IMDb ratings.

7.9.3 Rating vs. Number of Votes

A scatter plot was used to examine whether audience engagement, represented by the number of votes, is associated with movie ratings.

```
plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=model_df,
    x='Votes',
    y='Rating',
    alpha=0.5
)

plt.title('Movie Rating vs. Number of Votes')
plt.xlabel('Number of Votes')
plt.ylabel('IMDb Rating')
plt.show()
```

7.9.4 Average Rating by Genre

Since some movies belong to multiple genres, the genre values were separated before calculating the average rating for each individual genre.

```
genre_df = model_df.copy()

genre_df['Genre'] = genre_df['Genre'].str.split(', ')

genre_df = genre_df.explode('Genre')

genre_ratings = (
    genre_df.groupby('Genre')['Rating']
    .mean()
    .sort_values(ascending=False)
)

Genre_ratings
```

7.9.5 Average Rating by Genre Visualization

The average rating for each genre was visualized using a horizontal bar chart.

```
plt.figure(figsize=(10, 7))

genre_ratings.sort_values().plot(kind='barh')

plt.title('Average Movie Rating by Genre')
plt.xlabel('Average Rating')
plt.ylabel('Genre')
plt.show()
```

The analysis showed that Film-Noir had the highest average rating at approximately 7.68, while Horror had the lowest average rating at approximately 5.86 in the dataset.

7.9.6 Correlation Analysis

Correlation analysis was performed to examine the relationships between numerical movie features and the target rating.

```
correlation = model_df[
    ['Year', 'Duration (min)', 'Metascore', 'Votes', 'Review Count', 'Rating']
].corr()
```

correlation

The results showed that Metascore had the strongest positive correlation with IMDb Rating at approximately 0.58, followed by Votes at approximately 0.37 and Duration at approximately 0.36. Year showed almost no linear relationship with Rating.

7.9.7 Correlation Heatmap

A correlation heatmap was created to visually represent the relationships among numerical features and movie ratings.

```
plt.figure(figsize=(9, 6))
```

```
sns.heatmap(  
    correlation,  
    annot=True,  
    cmap='coolwarm',  
    fmt='.2f'  
)
```

```
plt.title('Correlation Heatmap of Numerical Features')  
plt.show()
```

The heatmap makes it easier to identify stronger and weaker relationships between the numerical features used in the analysis.

7.10 Preparing Data for Machine Learning

The processed dataset was divided into input features and the target variable. The data was then split into training and testing sets, with 80% used for training and 20% used for testing.

```
X = model_df[features]  
y = model_df[target]
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.2,  
    random_state=42  
)
```

```
print("Training data shape:", X_train.shape)  
print("Testing data shape:", X_test.shape)
```

The resulting training dataset contained 7,676 records, while the testing dataset contained 1,920 records.

7.11 Feature Preprocessing

Numerical features were standardized using "StandardScaler", while categorical features were converted into numerical form using one-hot encoding.

```
preprocessor = ColumnTransformer(  
    transformers=[  
        ('num', StandardScaler(), numeric_features),  
        ('cat', OneHotEncoder(handle_unknown='ignore'), categorical_features)  
    ]  
)
```

7.12 Model Building and Evaluation

Three regression models were developed and evaluated: Linear Regression, Decision Tree Regression, and Random Forest Regression. Each model was trained using the same preprocessing pipeline and evaluated using MAE, RMSE, and R² Score.

The final model comparison showed that Random Forest Regression achieved the best overall performance.

7.13 Actual vs. Predicted Analysis

The predictions generated by the Random Forest model were compared with the actual IMDb ratings.

```
prediction_comparison = pd.DataFrame({  
    'Actual Rating': y_test.values,  
    'Predicted Rating': y_pred_rf  
})
```

```
prediction_comparison.head(10)
```

This comparison was used to examine how closely the predicted ratings matched the actual ratings.

8. Data Visualization

Data visualization was performed to understand the distribution, patterns, and relationships present in the movie dataset. The visualizations helped identify how movie ratings are distributed and how different features are related to IMDb ratings.

8.1 Distribution of Movie Ratings

The distribution of IMDb movie ratings was visualized using a histogram with a density curve. This visualization shows how frequently different rating values occur in the dataset.

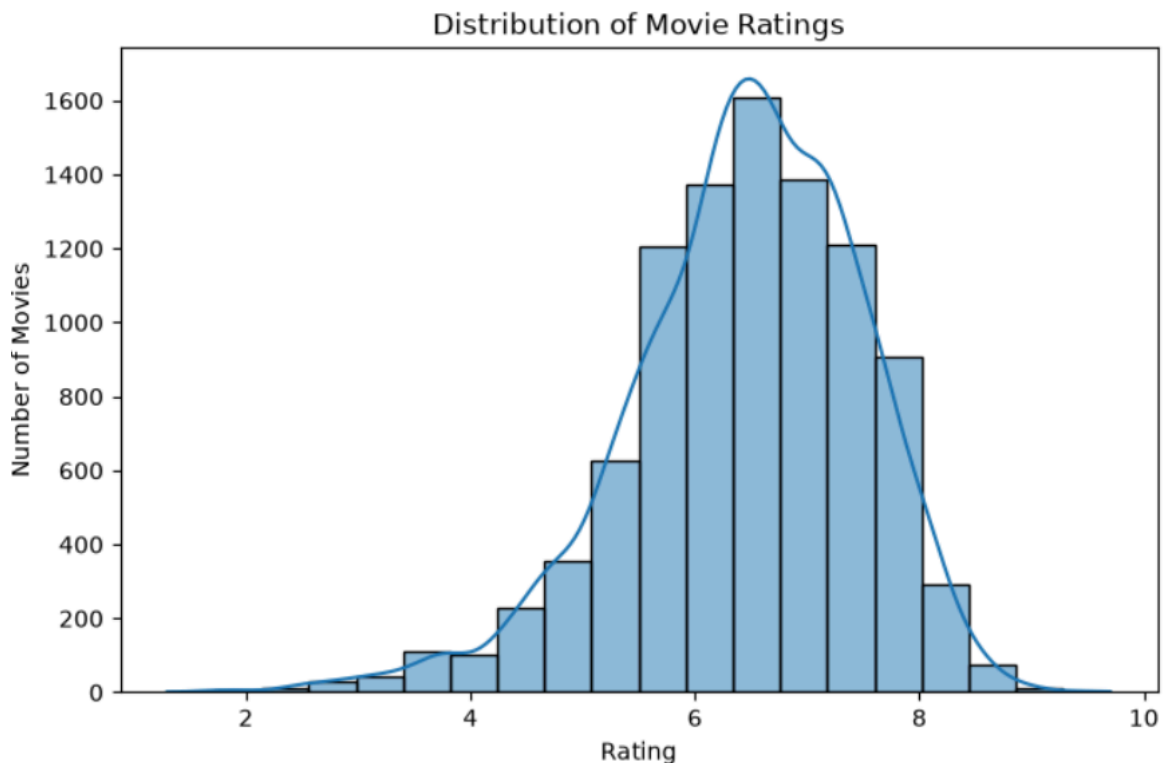


Figure 1: Distribution of Movie Ratings

The visualization shows that most movies have ratings concentrated around the middle-to-higher rating range, while very low and very high ratings occur less frequently.

8.2 Movie Rating vs. Metascore

A scatter plot was used to examine the relationship between IMDb movie ratings and Metascores.

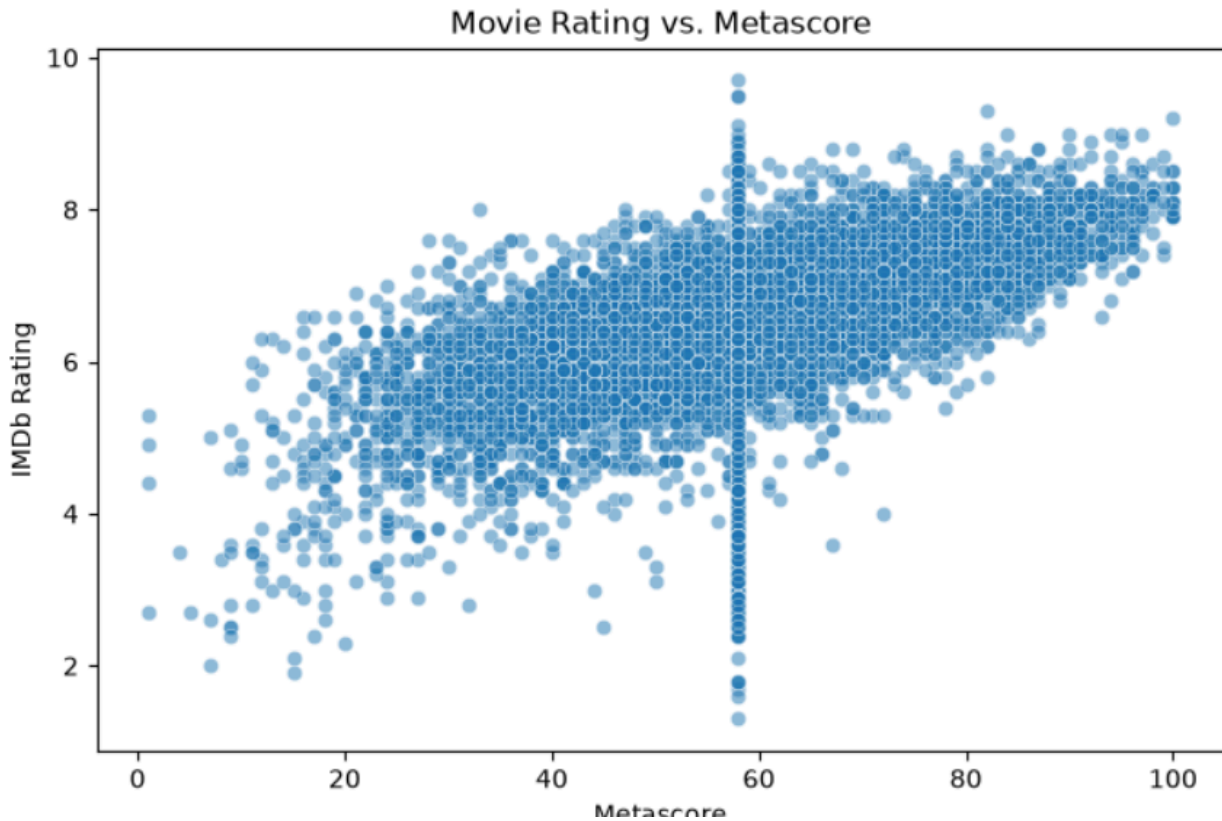


Figure 2: Movie Rating vs. Metascore

The visualization shows a positive relationship between Metascore and IMDb Rating. Movies with higher Metascores generally tend to have higher IMDb ratings, although the relationship is not perfectly linear.

8.3 Movie Rating vs. Number of Votes

A scatter plot was used to analyze the relationship between IMDb movie ratings and the number of votes received by each movie.

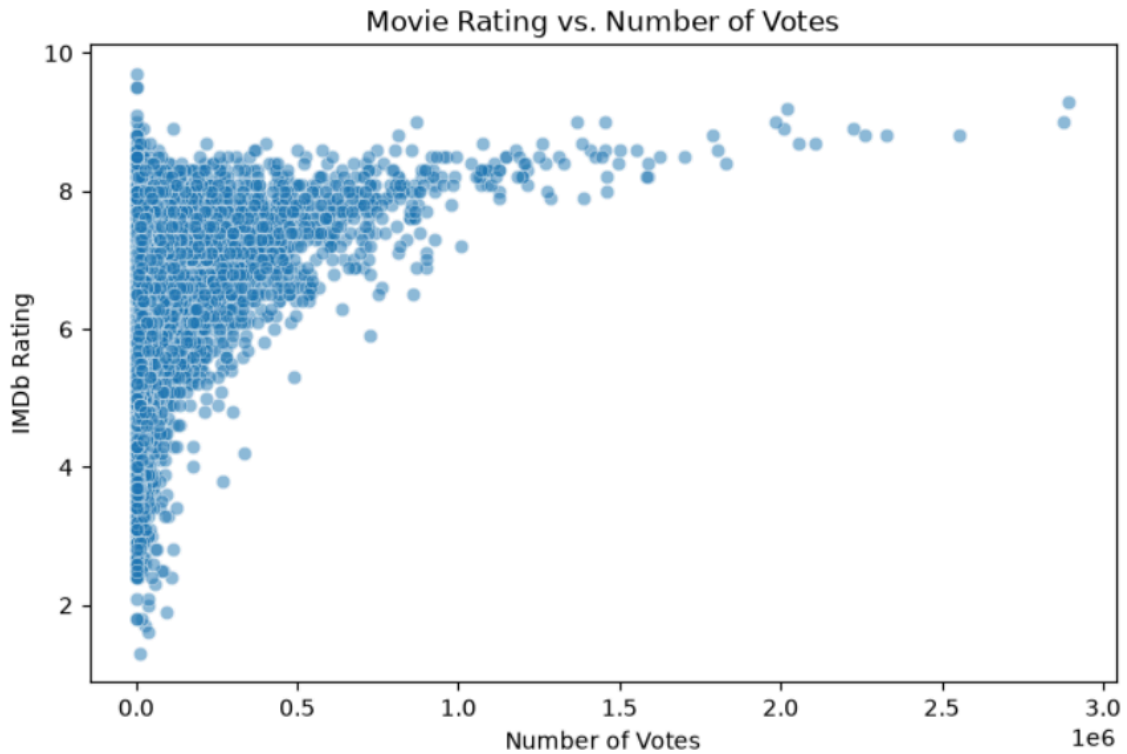


Figure 3: Movie Rating vs. Number of Votes

The visualization indicates a generally positive relationship between the number of votes and movie ratings. However, the relationship is not completely linear, as movies with different numbers of votes can have similar ratings.

8.4 Average Movie Rating by Genre

A horizontal bar chart was created to compare the average IMDb rating across different movie genres.



Figure 4: Average Movie Rating by Genre

The visualization shows noticeable differences in average ratings among genres.

Film-Noir achieved the highest average rating of approximately **7.68**, while **Horror** had the lowest average rating of approximately **5.86** in the dataset.

8.5 Correlation Analysis

A correlation heatmap was created to examine the relationships between numerical movie features and IMDb ratings.

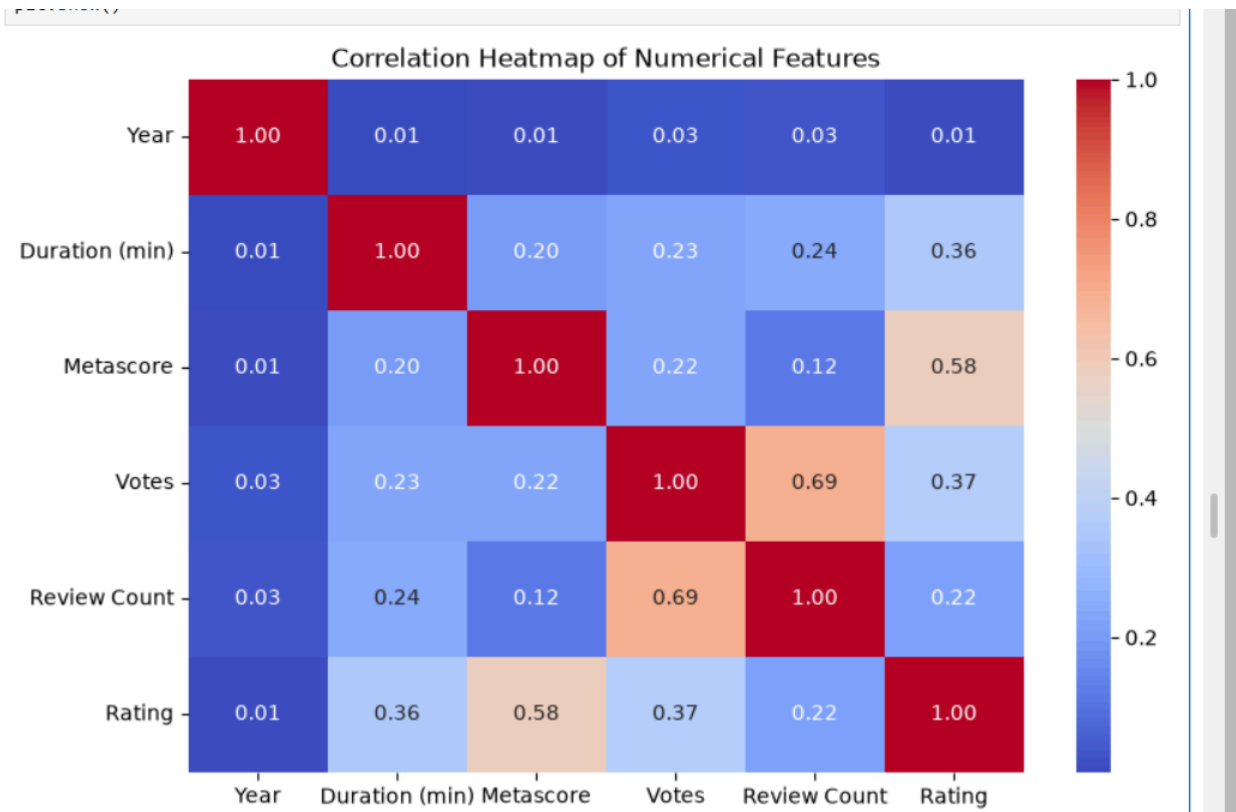


Figure 5: Correlation Heatmap of Numerical Features

The heatmap shows that Metascore has the strongest positive correlation with IMDb Rating, with a correlation of approximately 0.58. Votes and Duration also show moderate positive relationships with ratings, while Year has almost no linear relationship with IMDb Rating.

9. Model Results and Comparison

Three regression models were developed to predict IMDb movie ratings: Linear Regression, Decision Tree Regression, and Random Forest Regression. The models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

9.1 Model Performance Comparison

The performance of the three models is summarized below:

Model	MAE	RMSE	R^2 Score
Linear Regression	0.547	0.746	0.489
Decision Tree Regression	0.565	0.781	0.440
Random Forest Regression	0.512	0.709	0.539

Lower **MAE** and **RMSE** values indicate better prediction performance, while a higher **R^2 Score** indicates that the model explains a greater proportion of the variation in movie ratings.

Based on all three evaluation metrics, **Random Forest Regression achieved the best overall performance** among the tested models.

9.2 R^2 Score Comparison

The R^2 Scores of the three regression models were visualized using a bar chart to provide a clear comparison of their predictive performance.

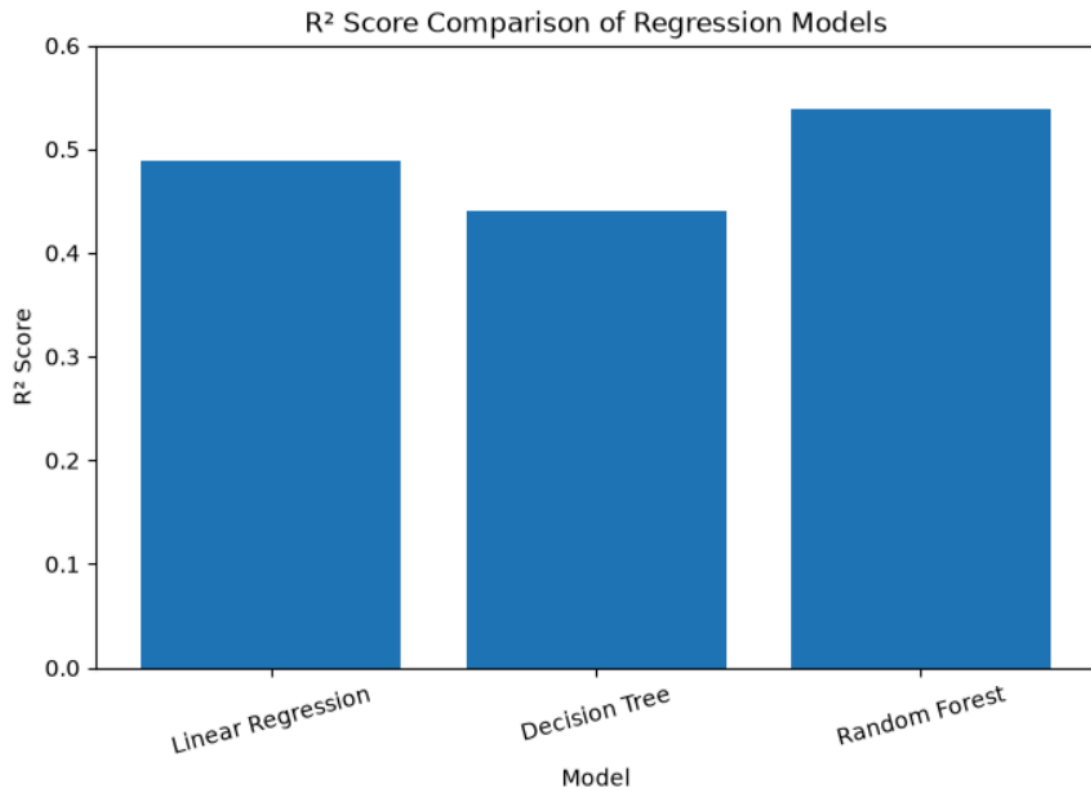


Figure 6: R² Score Comparison of Regression Models

The graph shows that **Random Forest Regression achieved the highest R² Score of 0.539**, followed by Linear Regression with 0.489. Decision Tree Regression achieved the lowest R² Score of 0.440.

Therefore, Random Forest Regression provided the strongest overall performance among the three tested models.

9.3 Actual vs. Predicted Movie Ratings

The actual IMDb ratings were compared with the ratings predicted by the Random Forest Regression model using a scatter plot.

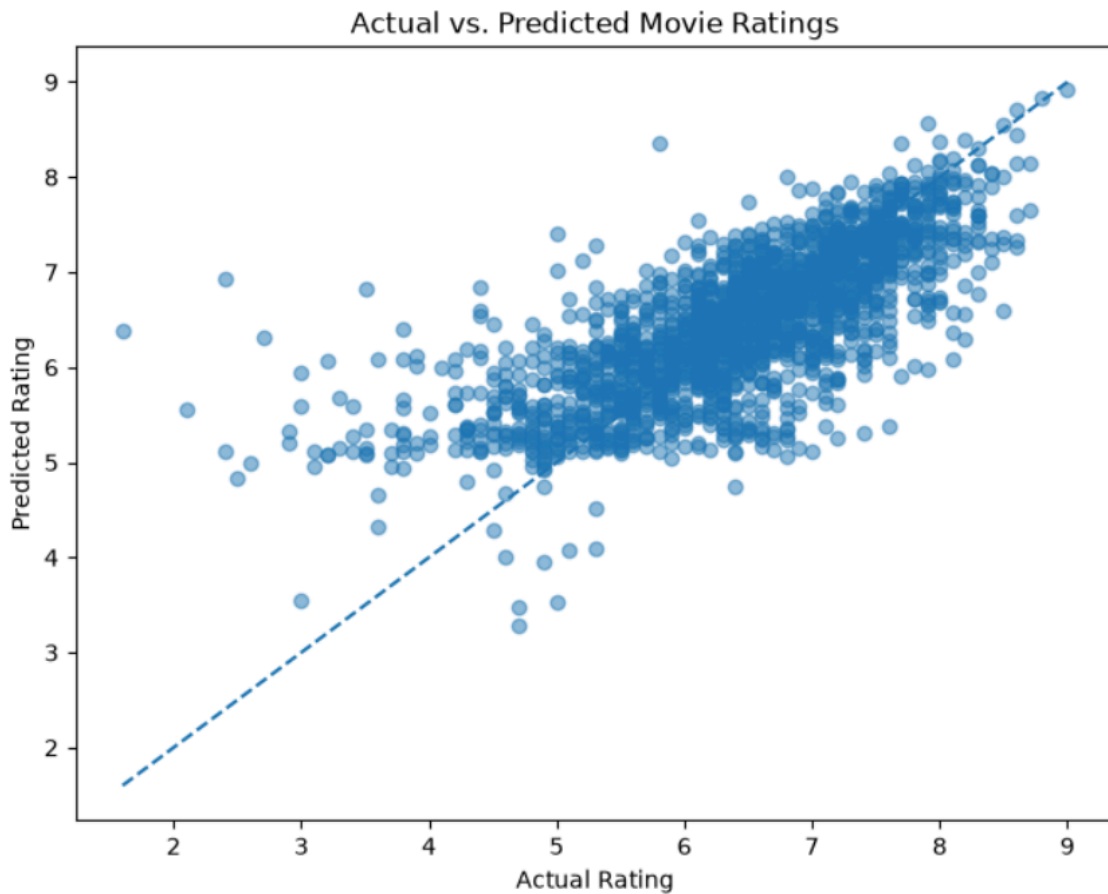


Figure 7: Actual vs. Predicted Movie Ratings

The diagonal reference line represents perfect predictions, where the predicted rating is equal to the actual rating. Points closer to this line indicate more accurate predictions.

The plot shows that the Random Forest model is able to capture the general pattern of movie ratings, although some differences between actual and predicted values are present.

10. Results & Evaluation

The developed regression models were evaluated to determine how accurately they could predict IMDb movie ratings. Three models were tested: Linear Regression, Decision Tree Regression, and Random Forest Regression.

The models were evaluated using three performance metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

10.1 Model Performance Comparison

Random Forest Regression achieved the lowest MAE and RMSE and the highest R^2 Score. Therefore, it was the best-performing model among the three tested models.

10.2 Evaluation Metrics

Mean Absolute Error (MAE): Measures the average absolute difference between actual and predicted ratings. A lower value indicates better performance.

Root Mean Squared Error (RMSE): Measures the square root of the average squared prediction errors. Lower values indicate better prediction accuracy.

R^2 Score: Measures how much of the variation in movie ratings is explained by the model. A higher value indicates better explanatory performance.

The Random Forest model achieved an MAE of **0.512**, meaning its predictions were, on average, approximately 0.51 rating points away from the actual ratings. Its R^2 Score of **0.539** indicates that the model explains approximately **53.9% of the variation** in movie ratings.

Key Findings

- Random Forest achieved the lowest MAE of **0.512**, indicating the smallest average prediction error.
- Random Forest achieved the lowest RMSE of **0.709**, showing better overall prediction accuracy.
- Random Forest achieved the highest R^2 Score of **0.539**, explaining approximately **53.9% of the variation** in movie ratings.

- Linear Regression performed better than the Decision Tree model on this dataset.
- Metascore showed the strongest numerical correlation with IMDb Rating, with a correlation of approximately **0.58**.
- Genre analysis showed that **Film-Noir** had the highest average rating, while **Horror** had the lowest average rating in this dataset.

11. Conclusion

This project demonstrated the use of machine learning regression techniques to predict IMDb movie ratings. The dataset was cleaned and prepared by handling missing values, converting text-based numerical fields, selecting relevant features, and applying appropriate preprocessing techniques.

Three regression models were developed and evaluated: Linear Regression, Decision Tree Regression, and Random Forest Regression. Among the tested models, **Random Forest Regression achieved the best overall performance**, with an MAE of **0.512**, an RMSE of **0.709**, and an R^2 Score of **0.539**.

The exploratory analysis also showed that **Metascore had the strongest numerical relationship with IMDb Rating**, with a correlation of approximately **0.58**. Genre analysis showed that Film-Noir had the highest average rating, while Horror had the lowest average rating in the dataset.

Overall, the project demonstrates how data preprocessing, exploratory data analysis, and machine learning can be combined to build a movie rating prediction system. The results also show that movie ratings are influenced by multiple factors, and the selected features alone cannot completely explain rating variations.

12. Future Improvements

The performance of the movie rating prediction system can be improved through the following approaches:

- Apply hyperparameter tuning to optimize the regression models.
- Use log transformations for highly skewed features such as Votes and Review Count.
- Improve genre representation by encoding individual genres instead of treating combined genre labels as single categories.
- Include additional movie features such as Director, Cast, and Certificate details.
- Apply advanced ensemble models such as Gradient Boosting or XGBoost.
- Perform feature importance analysis to identify the most influential factors affecting movie ratings.
- Increase the dataset size and use more diverse movie information to improve model generalization.

REFERENCES

1. Kaggle.

[IMDB Movies Dataset](#)

2. Python Software Foundation. Python Documentation.

[Python documentation](#)

3. Pandas Documentation. Pandas User Guide.

[Pandas Documentation](#)

4. Scikit-learn. Machine Learning in Python.

[Scikit-learn Documentation](#)

5. Matplotlib. Matplotlib Documentation.

[Matplotlib Documentation](#)